MomChef Homemade Bites

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Software Investigation and Analysis

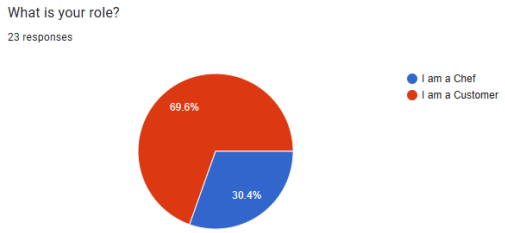
Conduct a software investigation to understand the requirements for the MomChef Homemade Bites App by gathering requirements using two techniques (with evidence), analyzing current systems, and identifying key functionalities needed.

A. Method 1: Survey

From the survey, I gathered valuable information's from 23 responses that support the needs of MomChef App from chefs and users. The people who filled the survey were chosen carefully. I gave the survey link to a chef who gave it to his chef's friends. Also, I gave the survey for users who order food frequently.

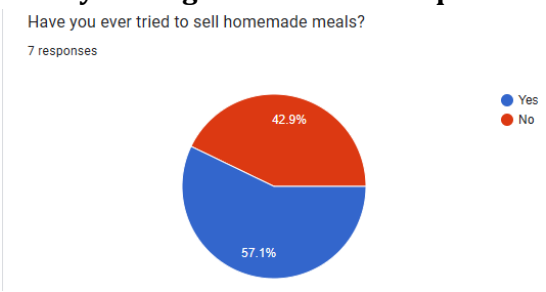
The survey helped me to understand user needs and requirements, and it included a carefully chosen questions that support what features should be implemented in the application.

Survey Link: https://forms.gle/8bbuJoDQ8PPi3byc7

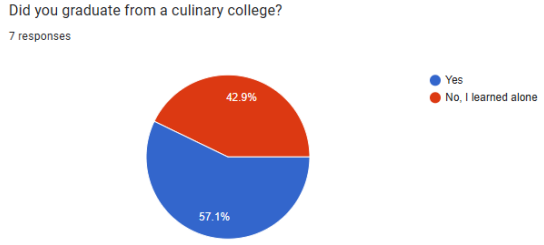


The chart shows that from the 23 responses 30.4% of them were chefs and 69.6% were customers.

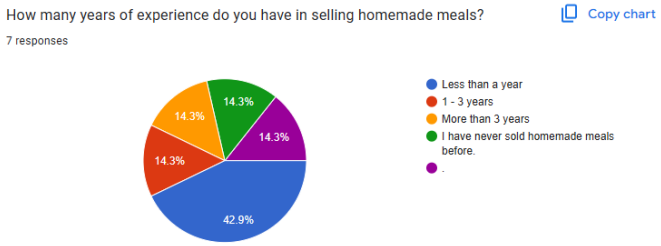
Survey findings based on chefs' questions and answers:



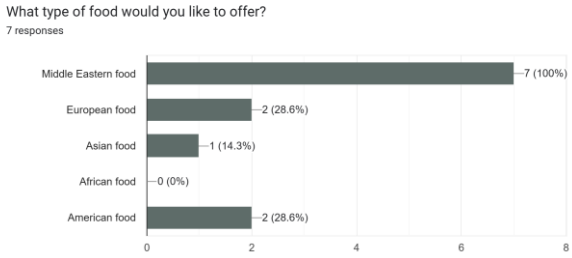
The chart shows that 42.9% did not sell homemade food and 57.1% tried to sell homemade food, which means that some of the chefs have experience.



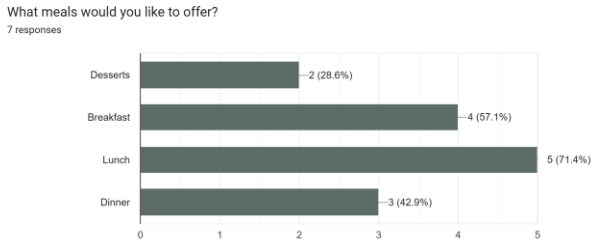
This shows that 57.1% of the chefs have professional experience in cooking food and 42.9% learned cooking alone without a professional experience, which means we can have all the chefs in the application.



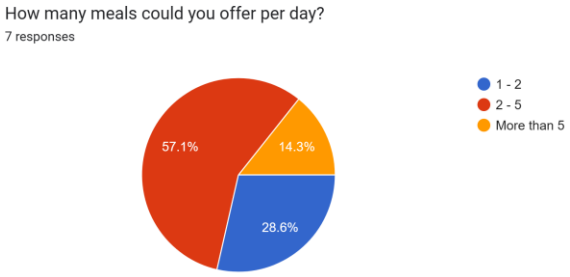
The chart shows that most of these chefs have a very good experience in selling homemade food.



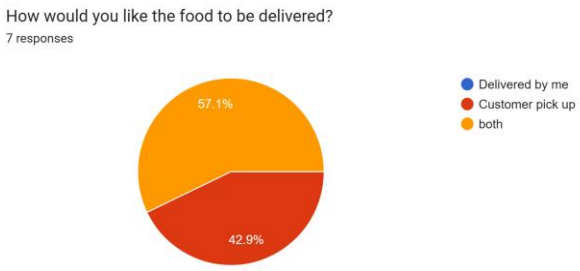
This chart shows that these chefs can offer multiple types of foods but no one offer African food, which means most types of food will be available.



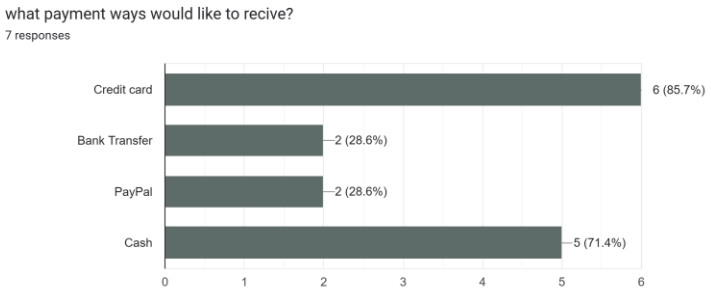
This shows that the chefs offer different meals, some of them are specialized in one aspect but some of them are specialized in different aspects, which means most types of food will be available.



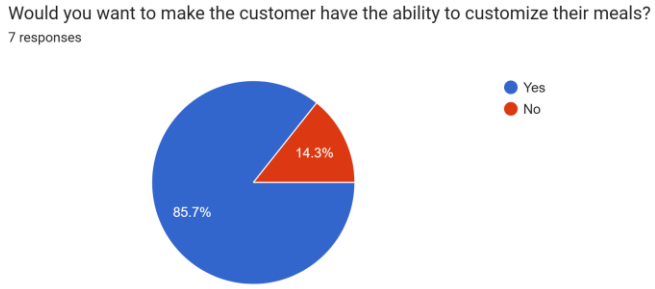
Most of these chefs can offer 2 – 5 meals per day which is good, and they are 57.1%, and 14.3% can offer more than 5 meals, and 28.6% can offer 1 – 2 meals.



The chart shows that 42.9% of chefs prefer customer pick up, but 57.1% can deliver the food by them, which means we could make the customer choose the delivery type.



This shows that chefs can receive payment in multiple ways, which means I must integrate all types of paying in the application.



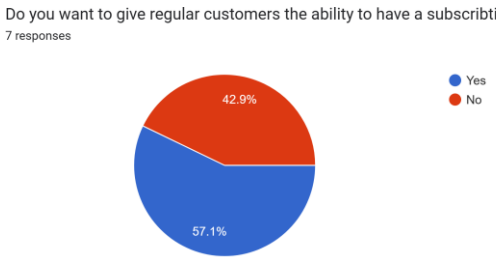
Most of the chefs want the customer to customize their meals, which means we will have a section for customers to customize the meals or add notes.



This shows that all of the chefs want a profile for themselves, so what we will do is to make profile for every registered chef.



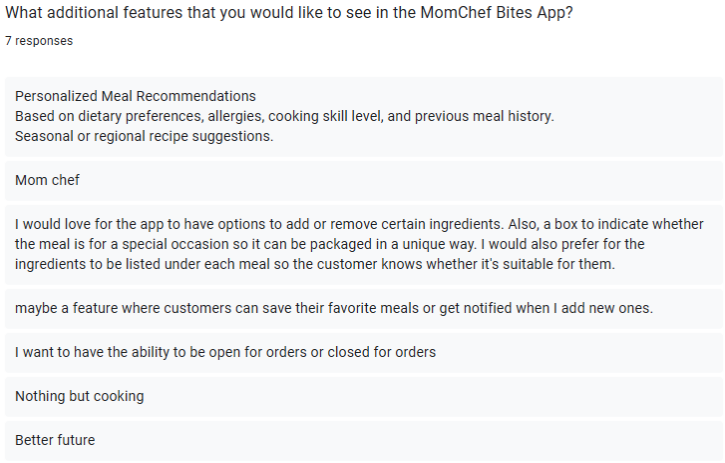
All chefs want to see rating, so we will add a section for rating meals.



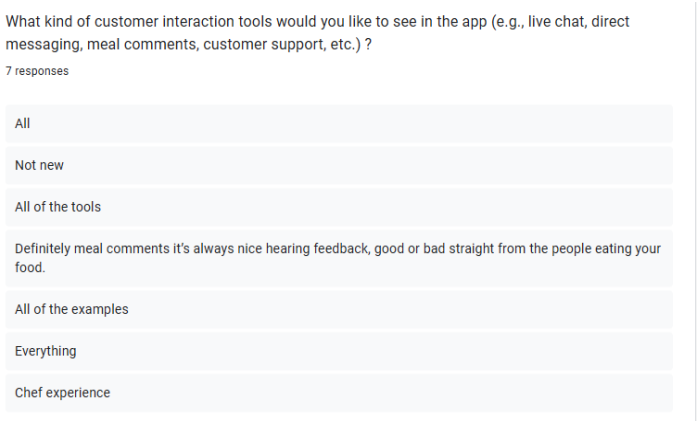
57.1% of chefs want to have subscription service, and 42.9% don't, we will see if we will integrate this feature in the application.



Most of them want to set minimum order price, so we will put a minimum order price for customers.

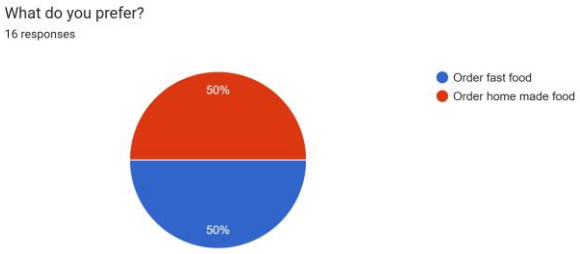


This question is to recognize additional features in the application, some of the suggestions were good, and some of them were not good. I will list the selected suggestions in the Requirements section.

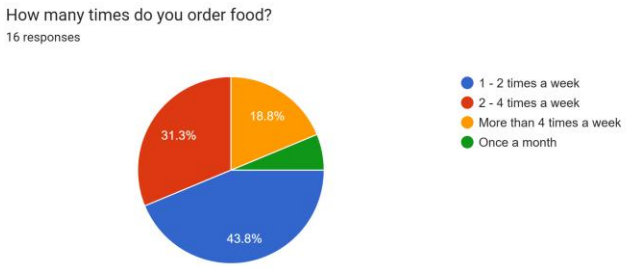


This shows that live chat, direct messages, meal comments, and customer support must be integrated into the application.

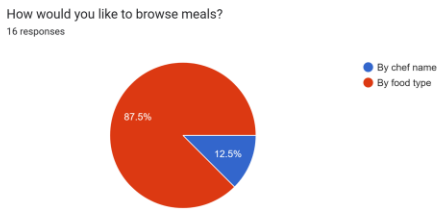
Survey findings based on chefs' questions and answers:



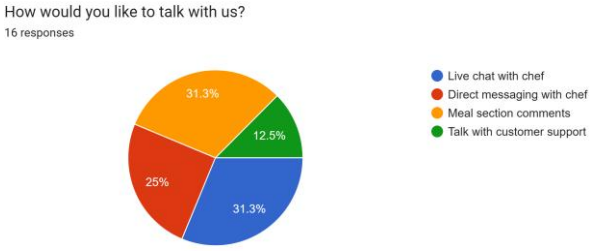
This shows that the 50% of customers order fast food and 50% of customers order home made food, which is good, we now know that we have customers.



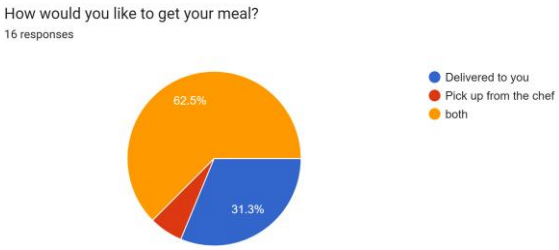
Most of the customers (75.1%) order from 1 – 4 times a week, and 18.8% order more than 4 times a week, and only 6.3% order once a month.



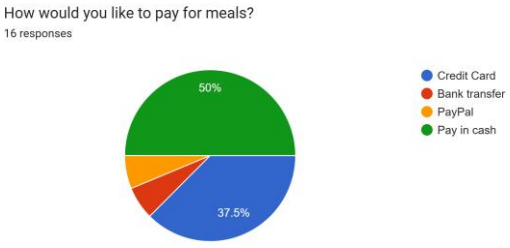
This shows that most of the customers like to search food based on food type, but the chefs said that they want dedicated profile, so we will see what's the best.



This shows that all of them should be integrated into the application, it's the same question that chefs answered before. But maybe we will not integrate customer support because only 12.5% want it.



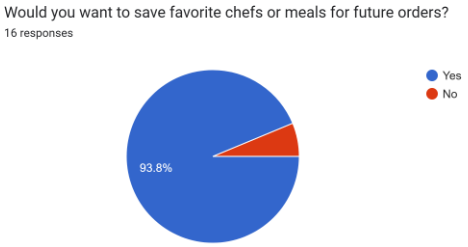
This shows that there should be an option to choose the delivery type.



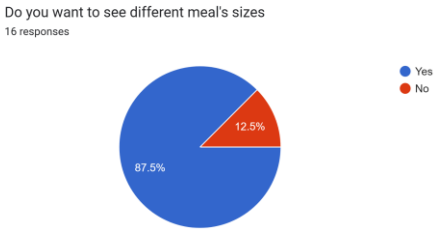
This shows that most of customer prefer to pay in credit card or cash, which means we might only have credit card and cash.



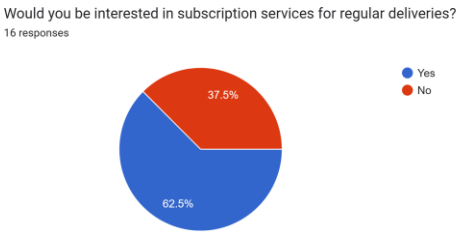
Most of the customers think that it is very important to track their order, which is a feature that will be implemented in the application.



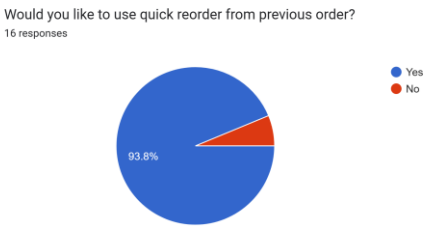
This shows that most of the customers want to save their favourite chefs.



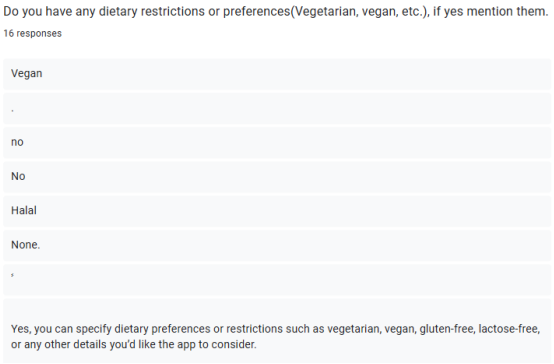
This shows that there should be different meals sizes.



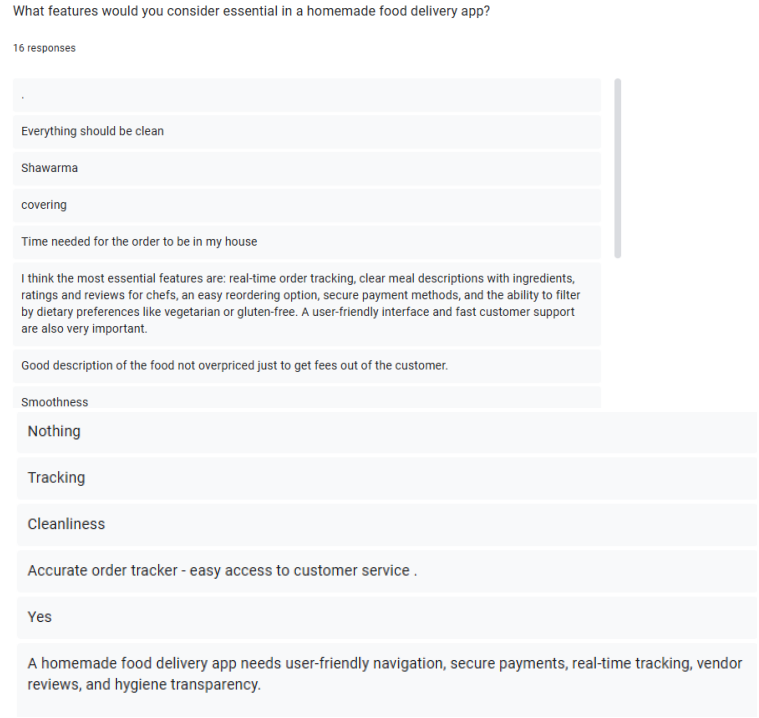
Some of them are interested in a subscription, and 37.5% of the customers were not interested.



This shows that there should be a feature to reorder a previous order with one click.



This shows that there should be dietary restrictions.



This shows that there are some suggestions that are good. I will list the good chosen suggestions in the requirements section.

B. Method 2: Interview

For the interviews, I made two interviews, one with a chef and one with a customer who worked before as a cashier for a restaurant. I gathered valuable information from the interviews that helped me gather information and transform them to requirements.

محمد: ممكن تعرفينا بنفسك؟

ديما: انا ديما عمري 24 سنة دارسة تكنولوجيا حيوية, وكثير بحب المطبخ, ودخلت المطبخ وانا عمري 14 سنة, كثير بحب اعمل حلويات وبحب اشوف الناس يوكلو اشي انا عملته ويعطوني ردة فعل حلوة.

محمد: هل بتحبي تبلشي تبيعي وجبات منزلية وشو الطريقة (سوشل ميديا, تواصي)

ديما: اكيد بحب ابلش وطبعا بفضل يكون تطبيق كثير احسن من السوشل ميديا او التواصي بلاش ما يصير خربطة بالاوردرات, التطبيق بكون كل اشي واضح.
محمد: كيف بتحبي يكون تصميم التطبيق؟
ديما: شخصيا بفضل يكون الوجبات موجودة واسم كل شيف تحتها وبناء عليه بختار.
محمد: لو ضفنا نظام يكون في بروفايل لكل شيف والوجبات الي ببيعها, شو بتفضلي يكون موجود بالبروفايل؟
ديما: وصف عن الشيف, أنواع المطابخ الي بتعامل معها, تقييم للشيف ووجباته, كم مدة الطبق لحتى يوصل.
محمد: بتحبي يكون عندك كبسة تفتحي او تسكري بروفايلك باي وقت او يكون ساعات عمل معين؟
ديما: عدد ساعات معين ليكون مثل نظام دوام.
محمد: هل بتحبي يكون في تقارير عن مبيعاتك؟
ديما: اكيد, بقدر اشوف ايش الأشياء الي بتنباع اكثر, واشوف البيست سيلر, وايش الناس ما بتشتري.
محمد: بتحبي يكون في نظام اشعارات لتقييم الوجبات والتعليقات الي بتوصلك؟
ديما: اكيد عشان اضل على تواصل دائم وما يضيع أي تقييم او كومنت عشان اوخذها بعين الاعتبار.
محمد: بتحبي يطلعلك إنذارات في حال تاخرتي على تجهيز طلب معين؟
ديما: حبيت فكرة الإنذارات عشان اقدر التزم بالوقت وما اتاخر على الزبون.
محمد: بتحبي يطلعلك اشعار انو في طلب اجاكي وتعمليله قبول او رفض وانتي برا التطبيق, او جوا التطبيق؟
ديما: لا بحب يجيني اشعار برا التطبيق ومن جوا.
محمد: بتحبي يكون الك إمكانية تصنفي وجباتك ضمن البروفايل؟
ديما: اه طبعا لانه بسهل عالزبون ايش بحب يختار من الاكل الي بعمله.
محمد: بتحبي تحددي عدد الوجبات الي بتقدميها باليوم, او لما يبطل عندك وجبات تسكري بروفايلك؟
ديما: بحب ابلش وبس يبطل عندي وجبات يتسكر البروفايل.
محمد: بتحبي يكون في محادثة بين الزبون؟
ديما: اه طبعا وبفضل تكون دايركت مسج, لانه بحسس الزبون انه رايه مهم واني رح اوخذه بعين الاعتبار.
محمد: بتحبي الزباين يضيفو ملاحظات عالاوردر؟
ديما: طبعا اكيد ممكن حدا عنده حساسيه من اشي معين, ما بده مكون معين, او بده اضيف مكون معين وغيره.
محمد: هل بتفضلي يكون عندك صفحة خاصة الك لتديري اوردراتك؟
ديما: اكيد, لانه انا اقدر اشوف شو الاوردرز الي عندي, الي خلصتها, الي لازم اعملها.
محمد: بتحبي يكون في تقييم للزبون؟
ديما: حبيت الفكرة لانه زي ما الزبون اله حق انه يقيم الشيف, برضو الشيف حلو يقيمه.
محمد: شو بتفضلي يجيكي طرق دفع؟
ديما: كل طرق الدفع.
محمد: اذا طلب التغى بتحبي تعرفي السبب ونحط انه ما يقدر يلغي الطلب الا لما يحط السبب؟
ديما: فكرة كثير حلوة وممكن نعطيه ثواني معينه يلغيها بدون سبب وبعد هيك يصير يلغي مع سبب.
محمد: بتحبي انتي كشيف تضيفي عروض على وجباتك؟
ديما: اكيد نعم.
محمد: بتحبي يكون عندك تواصل مع ادمنز التطبيق؟
ديما: اكيد لانه همه بعرفو اكثر بتطبيقهم لو صار معي أي مشكلة.
محمد: بتحبي يكون في فيتشر انو بقدر زبون يضيف وجبة من وجباتك لوجباته المفضلة؟
ديما: حبيت اه طبعا بحب لانه الاشي بسهل عليه يدخل عالوجبة المفضله ويعمللها اوردر.
محمد: بتحبي تزودي التطبيق بمكونات الوجبة؟

ديما: طبعا ولازم يكون موجود ويكون كل المكونات واضحه.

Customer Interview Link, Questions, and Answers : [Interview with customer (Ahmad)]

محمد: هل بتفضل يكون في فيديوهات قصيرة لكل شيف وكيف بساعدك هاذ الاشي يخلي تتخذ قرار الطلب؟

احمد: اه اكيد عشان يبينلك شو بعمل وشو طريقته وكيف نظافته والمكان الي بطبخ فيه, واكيد بساعدني اتخذ قرار الطلب.

محمد: كيف بتحب تقيم الشيف؟(حسب سرعة التوصيل، جودة الاكل، النظافه)؟

احمد: بالنسبة الي جودة الاكل والنظافة, اما سرعة التوصيل الشيف ما اله دخل فيها لانه مش هو الي رح يوصل, وممكن ياثر الشيف بهاذ الاشي ونعتبره سوء إدارة منه.

محمد: كيف بتحب تتواصل معنا (شات مع الشيف مثلا)؟

احمد: ما بتوقع في شات مع الشيف بس لو في بكون افضل, وهي خدمة عملاء بتكون كالعادة والمفروض الشيف يراجع المحادثات الي بتخصه.

محمد: كيف بتفضل تختار الشيف غن طريق تقييمه مثلا؟

احمد: حسب الشهرة بصراحة, بتكون الناس مجربيته غيري وبتقدر تشوف التقييم.

محمد: هل بتحب يكون في عداد بوضح عدد الاشخاص الي طلبو الوجبة؟

احمد: لا مش ضروري, بس ممكن كل شيف يكون عنده بيست سيلر, لو بدك تجرب شيف جرب البيست سيلر عنده.

محمد: بتفضل يطلع التطبيق الوقت التقديري لتحضير وتوصيل الطلب قبل ما تطلب؟

احمد: اه طبعا, اشي كويس كثير لانه مرات بدك تطلب وانت مستعجل.

محمد: هل الاشعارات التذكيرية بوجباتك المفضله بكون مزعج او مفيد الك؟

احمد: بصراحة بالنسبة الي مزعج.

محمد: هل بتحس وجود نظام نقاط لكل طلب بشجعك تطلب بشكل اكثر؟

احمد: اه طبعا حلوة الشغلة بتشجعك توصي من التطبيق اكثر.

محمد: هل بتفضل يكون في تصنيفات للاكل؟

احمد: اه طبعا عشان تعرف تدور عالقائمة الي بدك إياها.

محمد: بتحب تدور عالاكل من خلال بروفايل للشيف وفي كل الوجبات الي بقدمها او يكون عندك سيرش عاسم الوجبة؟

احمد: كل شيف لحاله ووجباته لحالها الها تصنيفات.

محمد: شو رايك بوجود قائمة لا اريد ان اطلب منها مجددا؟

احمد: صراحه الواحد بحفظ لو طلع اوردر سيئ فما رح يرجع يوصي منه مرة ثانية.

محمد: هل بتشوف وجود دعم فني مفيد لحل المشاكل بشكل اسرع؟

احمد: اكيد, لو في مشكلة بالاكل لمين بدي اعترض, لازم يكون في دعم فني ويكون في تعويض.

محمد: هل بتحب تعمل مقارنة بين الوجبات؟

احمد: اه طبعا اشي كثير كويس بصراحه, مثلا عندك اكثر من وجبة فيها حار بس وجبة فيها صوص, وجبة فيها هالابينو, وجبة فيها شطة.

محمد: هل بتحب تعمل مقارنة بين الشيفات؟

احمد: اه, لانه في شيفات اختصاصها بختلف عن الثاني وعشان تعرف تصنيفاتهم.

محمد: هل بتحب الوجبات يكون الها وصف تفصيلي بالمكونات؟

احمد: اه لازم يكون كل التفاصيل.

محمد: شو رايك تقدر تحط نوت للاوردر؟

احمد: اه طبعا اشي جدا مهم.

محمد: هل بتحب يطلعلك تقييم للشيف لحال او للشيف والوجبة, او للوجبة لحال؟

احمد: الشيف تقييمه لحال والوجبة تقييمها لحال.
محمد: هل بتحب التطبيق يقترح عليك وجبات بوقت معين؟
احمد: اه ليش لا ما بتأثر وبتكون احسن من الاشعارات التذكيرية.
محمد: هل بتفضل الواجهة يكون فيها كثير معلومات او تكون بسيطة؟
احمد: لا تكون بسيطة يحطلي شو اختصاص الشيف وشو بقدم بصورة صغيرة ولو بدي معلومات اكثر بدخل عالبروفايل تاعه.
محمد: شو بتحب نضيف فيتشرز؟
احمد: تقييم للدلفري, عشان شركة الدلفري يشوفو مين مقصر ومين لا.

C. Requirements gathered from both methods

I gathered valuable information, I analysed this information, and I understood the requirements of my application. The survey helped me to know opinions from many different people with different preferences.

The informations I gathered turned into valuable requirements that my system will be build based on these requirements.

1. Users can create account.
2. Users can login to their account and logout.
3. Chef should provide specific details of himself.
4. Chefs can apply to sell meals
5. After accepting, Chefs can add meal to their profile with all details.
6. Chefs must specify whether a meal is a best seller or not.
7. Chefs must mention the components of the meal.
8. Chefs must mention type of meal (launch, dessert, breakfast.etc.)
9. Chefs must mention Category of meal (Arabian, American, etc)
10. Chefs must put the price of the meal.
11. Chefs must provide image of the meal.
12. Chefs can edit their information.
13. Chefs can update availability of a meal.
14. Chefs can delete a meal of his profile.
15. Chef can see active orders.
16. Chef can manage active orders.
17. Chef can see his completed orders history.
18. Customer can see chef’s profiles.
19. Customer can browse meals through chef profile.
20. Customer can add meal to his cart.
21. Customer cannot order from two chefs at the same time.
22. Customer can open cart and checkout.
23. Customer can track his active order.
24. Customer can see his completed order history.
25. Customer can edit his profile data.
26. Customer can pay in different payment types.
27. Customer must provide his address along with the usual data (email, Username, etc.).
28. Admin can see total users.

29. Admin can see total chefs.
30. Admin can see total customers.
31. Admin can preview chef's request to sell meals.
32. Admin can accept chef's request to sell meals.
33. Admin can see chef's profiles (shops) with all chef data and meals.

Create a feasibility report for the MomChef Homemade Bites App system.

A. Introduction:
We must do a feasibility study on any project, including the MomChef Homemade Application to make sure that everything will work well, and to see what prerequisites we need to implement this project into a real-world application within the expected time, cost, and resources.

B. Legal Feasibility: This study makes sure that the application is fully legal and the application follows the rules and laws of Jordan, the application must follow these steps:
   - Get the commercial registration from the Jordanian Ministry of Industry and Trade.
   - Get a license for the MomChef Homemade Bites App from my local municipality.
   - Register in sales tax at the Jordanian Income and Sales Tax Department.
   - Determine the location where the owner of the application lives.
   - Get a work statement from the Jordanian Ministry of Labor.
   - Register to the Jordanian Social Security Corporation
   - MomChef Application must follow the Personal Data Protection Law of Jordan.
   - The registered chefs should be licensed from the Jordanian Food and Drug Administration.
   - The registered chef's kitchens should be

C. Economic Feasibility: This study is to understand and identify the expected cost along with the expected benefits.
Project costs:

	Count	Unit Hours	Cost/Hour	Subtotals	Level 1 Totals	% of Total
1.0 Project Management					$187,000	42%
Project Manager	1	500	$100	$50,000		
Software Developers	4	1500	$50	$75,000		
Testers	2	700	$60	$42,000		
Designers	2	500	$40	$20,000		
Marketing Team	2	1000	$20	$20,000		
2.0 Hardware			Cost/Unit		$5,800	1%
Laptops	8		700	$5,600		
Cloud Server subscription/per year	1		200	$200		
3.0 Software					$132,500	30%
Front End Development		500	$125	$62,500		
Back-End Development		100	$125	$12,500		
Design		200	$100	$20,000		
Testing		300	$125	$37,500		
4.0 Marketing					$25,000	6%
Marketing expenses				$25,000		
5.0 Support						
Professional Support				$10,000	$20,000	
Maintenance				$10,000		
Reserves (20% of the budget)					$74,060.0	17%
Total Project Cost Estimate					$444,360	

Project Benefits:

I Evaluated the project's worth by making a Return of Investment.

The project starts developing at year 0 and it will take less than a year, so in the few months in year 0 after finishing the project, the expected benefits are $10,000.

	Year 0	Year 1	Year 2	Year 3	Total
Total Cost	$444,360	$20,000	$20,000	$20,000	$464,360
Total benefits	$10,000	$100,000	$200,000	$500,000	$810,000

ROI = (Total benefits – Total cost) / Total cost

= ($810,000 - $464,360) / $464,360

= 345,640 / 464,360

= 74.4%, this is the ROI in the first 3 years of launching the application.

D. Technical Feasibility: A study to assess all the resources we need to develop the MomChef App, and what we need is:

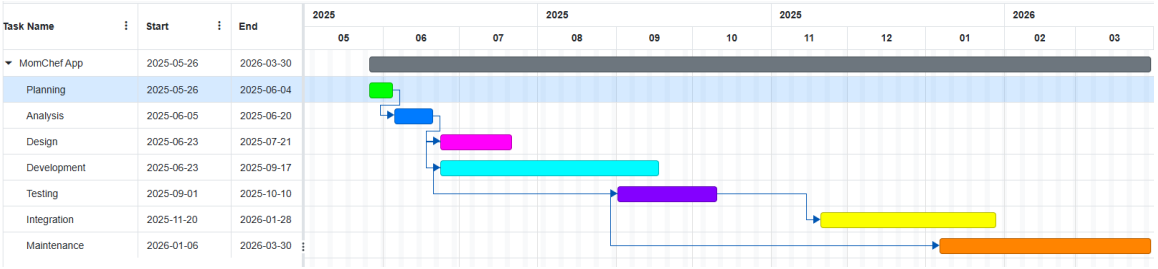
- Cloud Server: To host the website and the data base we need a cloud server such as (AWS)
- Front-end technologies. (HTML, CSS, JS, React)
- Back-end technologies. (PHP, Django)
- Open-Source IDE's. (VS CODE, Eclipse, etc.)
- Payment gateways. (PayPal)
- Laptops.
- Developers.

- Designers.
- Testers.
- Marketing team.
- Project Manager.

E. Operations Feasibility: A study to see how the app will run, and to see the operational needs of the Application.
   - The application will have three user roles (chefs, customers, and admin).
   - The application will be user friendly to make sure that users can easily use this application.
   - Chef's must have their own kitchen or space.
   - The application is scalable and there can be a lot of chefs and customers.
   - To make the application run on the long term, we need to frequently do maintenance, fix problems, and have a plan for long-term running.

   The application will have the usual operations such as planning, design, development, testing, and maintenance.

F. Scheduling Feasibility: A study to assess the timeline for the project to be completed. The project will take 8-9 months. Some of the tasks can start together, and some tasks can't start before the previous task finish.
   For example, when design start, development can start, and when planning finish, analysis can start. The project should be completed within the timeline provided.



G. Conclusion:
   In conclusion, and after we made the feasibility study, we made sure that the project can be implemented in real world, the application is fully legal, economically strong, technically and operationally strong, and manageable in schedule. In my opinion, and according to the feasibility study, the project will succeed.

Create a Software Requirements Specification (SRS) document for the MomChef Homemade Bites App system.

A. Project description:
   MomChef Homemade Bites is an application that helps chefs to start their business by selling homemade meals, the application was made to connect chefs with customers who are looking for homemade meals. Chefs will be able to manage their meals and orders, and the customers will be able to browse and order meals from chefs based on the preferences and ratings.

B. User requirements: There are three user roles (Chefs, Customers, Admin)
    1. User should be able to create account.
    2. User should be able to change profile data.
    3. Chef should be able to apply to sell meals.
    4. Chef should be able to categorize his meals.
    5. Chef should be able to specify the food type they sell.
    6. Chef should have a category in his profile for the best seller meal which will be the first in his profile.
    7. Customers should be able to browse meals.
    8. Customers should be able to order meals.
    9. Customers should be able to view order history.
    10. Customers should be able to track their order.
    11. Admin should be able to manage users.
    12. Admin should be able to accept chef applications.
    13. Admin should be able to see chef profiles
C. System requirements:
    1. System should allow users to manage their profiles.
    2. System should be compatible with multiple operating systems.
    3. System should secure confidential data.
    4. System should support English.
    5. System should be user-friendly and easy to use.
    6. System should support multiple payment methods.
    7. System should have minimum number of delays.
    8. System should secure payment data.
    9. System should allow customers to track order.
D. Functional Requirements:
    1. The system should allow users to register and login securely.
    2. The system should allow users to update their profile data including image, name, username, and password.
    3. The system should allow chef to create and manage meals.
    4. The system should allow chef to receive, and update orders.
    5. The system should tell users about order status.
    6. The system should provide tracking order functionality.
    7. The system should allow admins to view chef request and accept them.
    8. The system should allow chef to delete meal.
    9. The system should allow customers and chef to view order history.
    10. The system should allow customers to open their cart, manage order, and checkout.
E. Non-Functional requirements:
    1. Confidential data should be secured.
    2. There should have minimum number of delays.
    3. The application should be user-friendly.
    4. The application should support different screens.

5. The application should support different devices.
6. Only admins should access all the data except the confidential data.
7. The application should be scalable.
8. The application should support English.
9. Profile's images must be in a high quality.

Discuss the reason for selecting a particular lifecycle model for the development of your MomChef Homemade Bites App system.

The chosen lifecycle model for the MomChef Application is Agile, because Agile has an iterative behaviour and focuses on the collaboration that deliver each step and get feedback iteratively.

In MomChef project, Agile will work by breaking the project into small iterations, each iteration will be developed, tested, and presented before starting the next sprint, for example we have a iteration for the sign-up functionality, and we have a iteration for the sign-in functionality, the sign-up iteration will be developed, tested, and presented before we start in the sign-up iteration.

Agile is good for the MomChef Application because the requirements of an application like this are not fixed and fully clear and late changes in Agile model are welcomed, we could integrate a new requirement or remove a requirement from the application easily.

The interface must be user friendly, and the use of Agile guarantees that the interface is friendly by getting continuous customer feedback. Also, there is a high collaboration between teams and team members, everyone is included in the process including the customer which means that everything is clear, and the knowledge will be shared between the team, the team members, and the company.

As we said, Agile is an iterative approach, which means that always will be continuous testing within each iteration, this allows us to detect issues and risks early. For example, when developing the functionalities such as quick reorder functionality and tracking order functionality, we can address potential issues such as non-working quick reorder button, or low accuracy of tracking the order.

The Agile model includes continuous testing which make sure that all iterations of MomChef App are accurate and validated. By testing each step, we can identify failures, and issues early, which leads to more stable, and reliable application.

Agile Model work in MomChef project:

- The project is separated into smaller iterations, each iteration has specific tasks.
- The iteration will be planned, as we will define iterations aim and goals.
- The team will collaborate to work on each iteration, test them, and present them.
- The code will be shared with the iteration team members.
- The company will see and review each iteration, and gather feedback from them, making sure they are included in the process

In conclusion, Agile is the most suitable model for MomChef Homemade Bites Application to meet customer and user's expectations and due to the reasons, I provided.

Software Design Techniques and Behavioural Tools

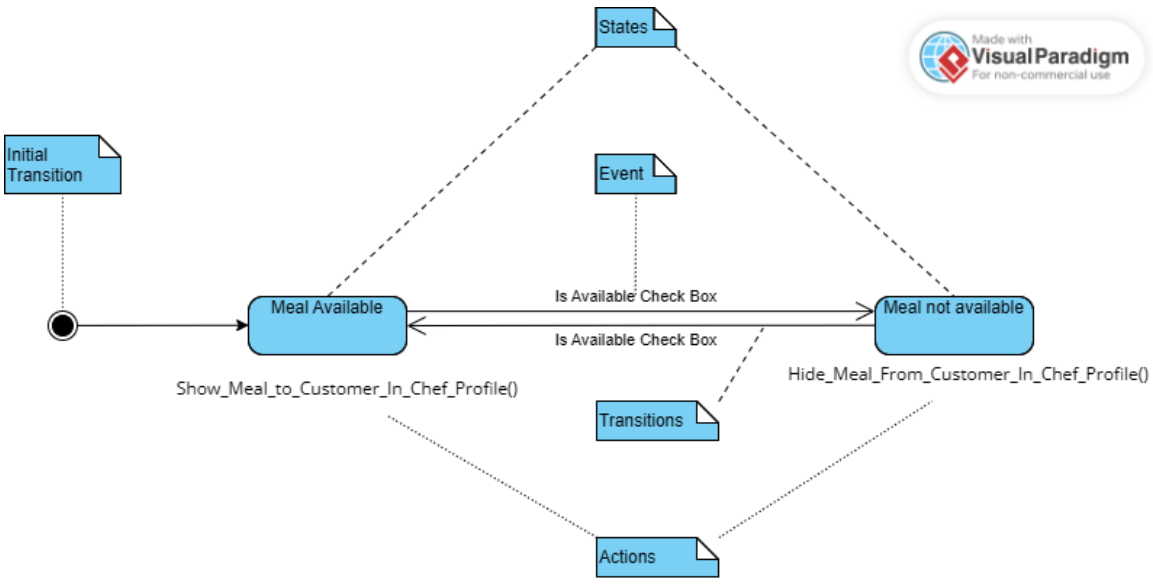
Finite state machine (FSM) and an extended FSM

FSM has states, and there is transitions between these states which Is a response to an event that could by any trigger. But an EFSM is and extension from FSM and it is the same as FSM but the key difference that EFSM transitions are represented in certain conditions to transfer from one state to another. EFSM is used to represent more complex systems that FSM is not applicable on that system, and FSM is used to represent less complex systems.

After discussing what is FSM and what is EFSM we should discuss the differences between them. As we said that FSM transitions are response to events such as pressing a button moves from one state to another like when user submit the login form it will log in the user, and when user press on logout button it will logout, but EFSM transitions are response to events and conditions like if-statements such as in order to make the chef update order status, that order must follow these conditions "is order checkout == true" true, meaning that the user pressed on checkout button, and "is order completed == false".

Another difference is that in FSM, the actions during the transitions are simple and do not have to update variables, while EFSM actions during the transitions are more complex and they may update variables or interact with other processes of the system.

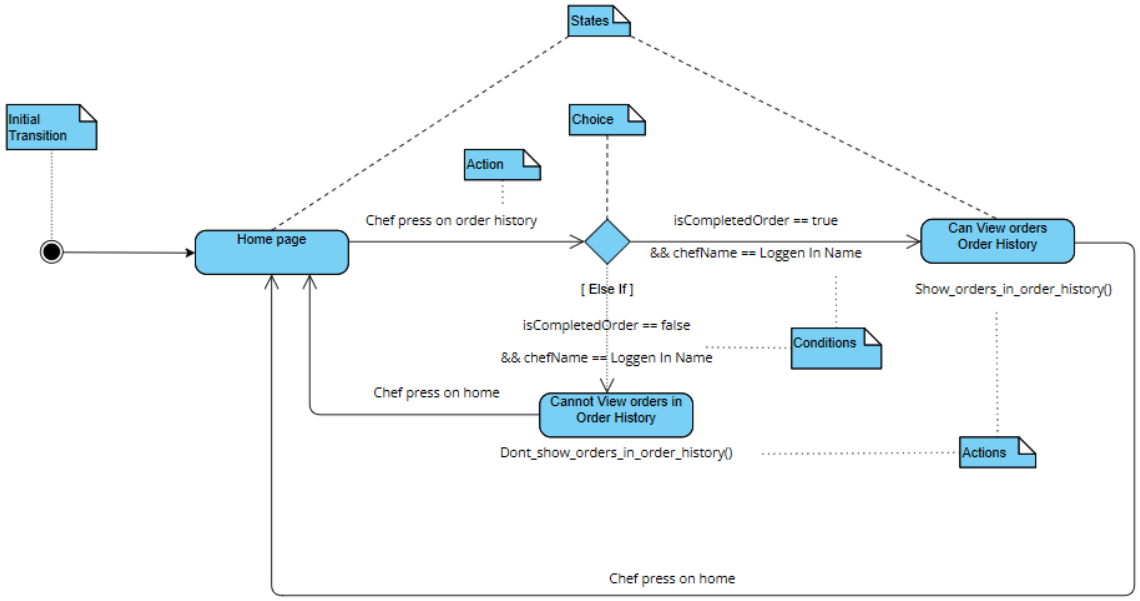
FSM example on the MomChef Homemade App:



This is an example that shows the object meal and its states which is "meal available" and "meal not available". The transition is when the chef presses on the check box that

transition from one state to another. When the state is “meal available” there will be an action which is showing the meal in chef profile for customers. When chef presses on the check box it will transition to the other state “meal not available” and do an action which is hiding the meal in chef profile for customers.

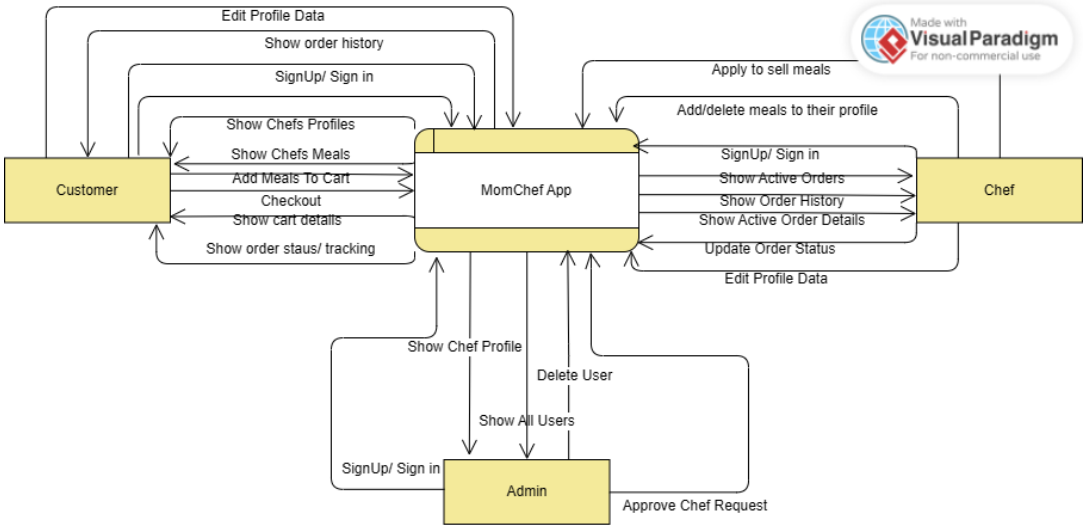
EFSM example on the MomChef Homemade App:



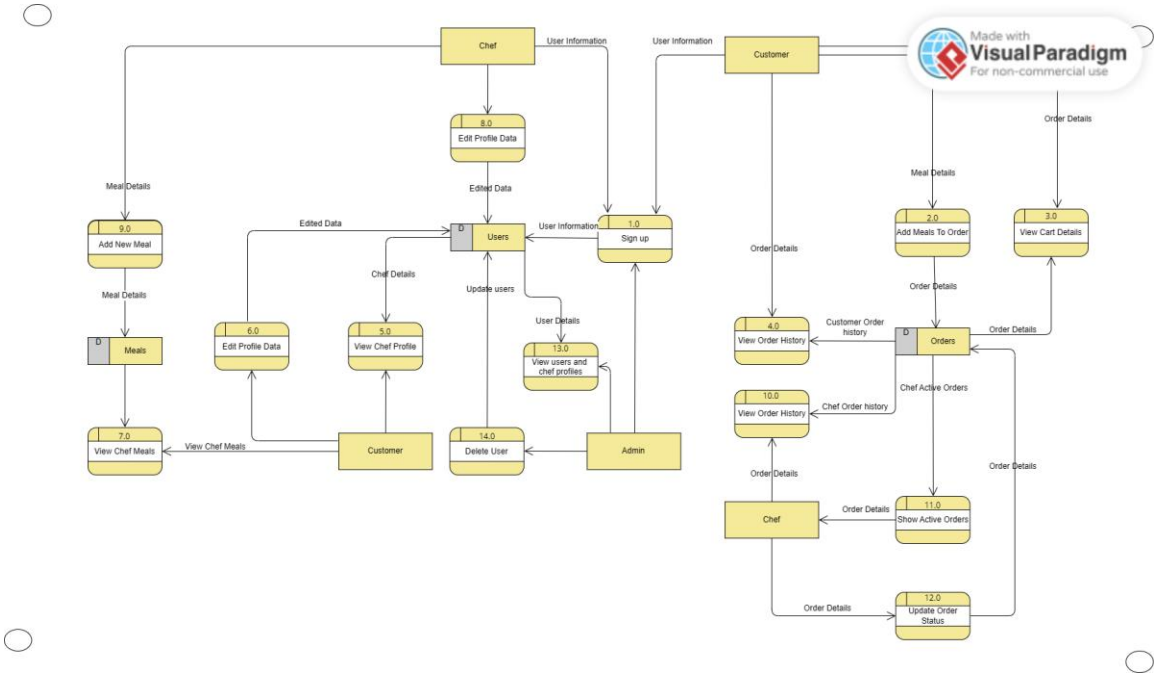
This is an example that shows the object Chef ability to view order history and its states which is “Home Page” as an initial transition, “Can view orders in order history”, and “Cannot view orders in order history”. The transition is when the chef presses on order history in the navigation bar, after that there is a guard conditions, so to transition to “Can view orders in order history” there are guard conditions that must met which is “Order should be completed” and “order chef name must be equal to the logged in chef name”, and to transition to “Cannot view orders in order history” there are guard conditions that must met which is “Order is not completed” and “order chef name must be equal to the logged in chef name”. Also, there are transitions that goes to the “home page” state, which is the chef pressing on the home button in the navigation bar.

Software Design Specification (SDS) document for the MomChef Homemade Bites App system.

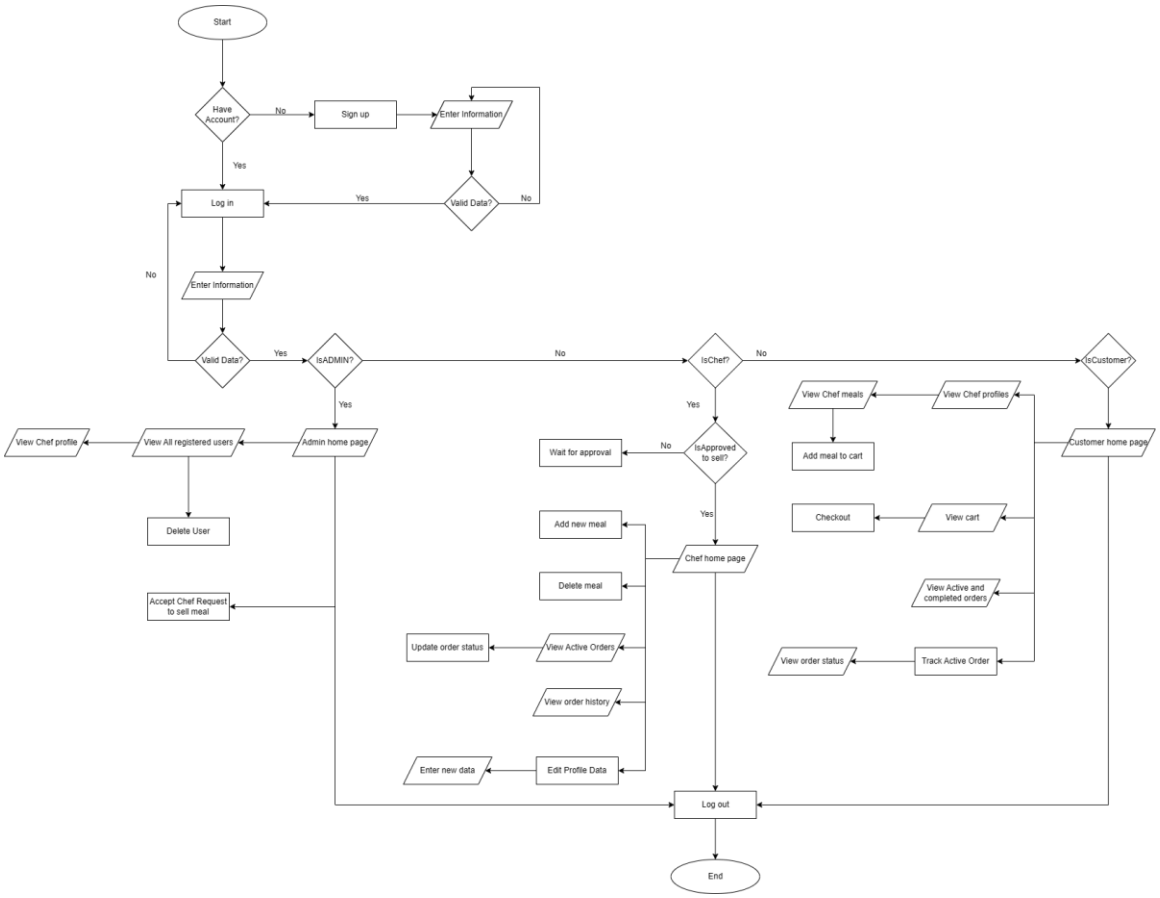
- DFD Context Level:



- DFD level 0:



- Flowchart:



Implementation of the MomChef Homemade Bites App system.

I implemented the application to support different screen sizes.

The Application Link: MomChef Adalo Application

Overview of the application: Adalo Overview - Link

Software Quality and Testing

Evaluating the process of undertaking a systems investigation with regard to its effectiveness in improving software quality.

In the system investigation for the MomChef App, I focused on collecting reliable, and accurate information to identify requirements that are applicable in real world applications and to make sure that everything is accurate and all requirements meet the needs of the chefs that will sell meals, and for the customers who will use the app, so I used two effective methods that made me understand the requirements for such system which they are survey and one interview with a chef, and another with a customer.

I used survey that collect information from both chefs and customers. The survey included great questions that helped me to identify the users' preferences. Because I used survey, I was able to collect valuable information from a large number of users.

Also, I used interviews, I made an interview with a chef, and interview with a customer. The interview with the chef included many questions that helps me understand what these chefs are looking for in the application. The interview with the customer helps me understand what these customers are looking for in the application.

All findings from these both methods were documented clearly just like the way I provided in part 2.1, I analysed the survey questions, and I listed them with their analysis, I also wrote all the questions and answers from the chef interview and from the customer interview. After that I decided to list all the requirements gathered from both methods, and the were clear and well-defined.

I had multiple objectives to achieve in my system investigation, I wanted to gather accurate and reliable information that make me understand the requirements and to gather applicable and achievable requirements. Also, I wanted to know the peak times for applications that are used for ordering meal, as I had a question in the survey "How many times do you order food a week?".

The system investigation helped me to identify new objectives for the MomChef App, I will list all objectives before and after the system investigation:

- Help talented home chefs to improve their selling rate.
- Make the ordering process for home meals easier.
- Help customers that order home meals frequently such as foreign students to view and order from a lot of professional chefs.
- Stakeholders' satisfaction rate is 90% and more.
- Get 95% successful order completion rate after three years.
- Get more than 50% ROI in the first three years.
- Complete the project within the time and budget.
- Maintain security of any sensitive data.

I had many evaluation criteria to know if the project is as successful as expected, these evaluation criteria were that implementing the whole fully functional software with three user roles with a good graphical user interface, the operating system operates smoothly and secure the sensitive data, stakeholders satisfaction with the whole application should be high, and get a high rate of ROI in the first three years. These evaluation criteria were taken into place in the process of building the application in every phase including design, development, etc.

After finishing the application, I showed the application for some stakeholder before deploying, to get their satisfaction rate, as it is one of the evaluation criteria. I involved the chefs, and customers in most of the stages of building the application, which helped me to

identify what the stakeholders really want from the application and what features they need in the application to make them satisfied. After completing the project, most of the stakeholders give me a positive reaction to the application from the sign up to order delivery as most of their requested requirements were implemented in the application.

The timeline and cost were taken into place, as I made the whole assessment of the timeline and cost in the feasibility study, as the involvement of stakeholders helped me to reduce the risk of misaligned features and the cost of changes. Also, planning was effective as I made a great Gantt chart and financial analysis based on the features and functionalities we implemented.

When I was working on the project, I always used to take feedback from stakeholders, and because the lifecycle model chosen was agile, it helped me a lot in getting feedback from stakeholders so there are many lessons learned throughout the journey, as I will always involve stakeholders at every stage because they are the key of success, well-defined and clear requirements and documentation always makes sure that everything is working as planned, and communication is a key to success, as it helps us understand everything.

Requirements Traceability Matrix (RTM) document for the MomChef Homemade Bites App (

RTM:

Requirement ID	Functional Requirement	Requirement Description	Test ID	Test Description	Test Design	Test Designer	UAT Test Req?	Test Env
Req01	Signup	Signup with already registered email	TC01	Signup with already registered email	Completed	Mohammad	No	Passed
Req01	Signup	Signup without filling any field	TC02	Signup without filling any field	Completed	Mohammad	No	Passed
Req02	Login	Login with non-valid data	TC03	Login with non-valid data	Completed	Mohammad	No	Passed
Req02	Login	Login without filling any field	TC04	Login without filling any field	Completed	Mohammad	No	Passed
Req03	Chef Create Meal	Create meal without providing a picture	TC05	Create meal without providing a picture	Completed	Mohammad	No	Passed
Req03	Chef Create Meal	Create meal without providing a name of the meal	TC06	Create meal without providing a name of the meal	Completed	Mohammad	No	Passed

Req04	Customer Checkout	Customer try to checkout with no meals in the cart	TC07	Customer try to checkout with no meals in the cart	Completed	Mohammad	Yes	Passed
Req04	Customer Checkout	Customer try to checkout with meals in the cart	TC08	Customer try to checkout with meals in the cart	Completed	Mohammad	No	Passed
Req05	Chef update order status	Chef update status to preparing	TC09	Chef update status to preparing	Completed	Mohammad	No	Passed
Req05	Chef update order status	Chef update status to completed	TC10	Chef update status to completed	Completed	Mohammad	No	Passed

Requirment ID	Functional Requirement	Requirement Description	Test ID	Status
Req01	Signup	Signup with a already registered email	TC01	Pass
Req01	Signup	Signup without filling any field	TC02	Pass
Req02	Login	Login with non-valid data	TC03	Pass
Req02	Login	Login without filling any field	TC04	Pass
Req03	Chef Create Meal	Create meal without providing a picture	TC05	Pass
Req03	Chef Create Meal	Create meal without providing a name of the meal	TC06	Pass
Req04	Customer Checkout	Customer try to checkout with no meals in the cart	TC07	Pass
Req04	Customer Checkout	Customer try to checkout with meals in the cart	TC08	Pass
Req05	Chef update order status	Chef update status to preparing	TC09	Pass

Req05	Chef update order status	Chef update status to completed	TC10	Pass

Evaluation and Justification

Assessing the merits of applying the Waterfall lifecycle model to a large software development project.

As we discussed before In Part 1.1, waterfall model is an sequential model, that works phase by phase, when a phase is completely finished, we can go to the next phase, so we must completely finish a phase before the next phase starts which would work well in large complex projects as everything should be 100% clear including requirements, and risk. Waterfall model also focuses on documentation, as it makes sure that all potential issues are identified, risk and risk response plan are clear which reduce the risk of any bad thing which is good for large projects to not spend any unexpected cost. In the waterfall model, all requirements should be clear at the start which makes us create a clear estimation of the timeline, milestones, resources and cost. Waterfall ensures that quality is high, but it may take higher costs or more time.

There is a software where the waterfall model was applied, which is the enterprise system of a AirRoyal airline company that Huub J. M. Ruël, Tanya Bondarouk, and Stefan Smink made case study of it. AirRoyal decided to work on the Enterprise system in a sequential stages.

In the enterprise system implementation, waterfall offer a very structured methodology as the project size required a methodology where phases are working in a sequential way, requirements are clear and well-defined, and all responsibilities are clear. The waterfall model prevented teams working on the project from skipping tasks to make sure that the progress is going well which is important to prevent errors in this large enterprise system.

This approach helped the company to have a strong documentation for the project, as all requirements, decisions, and risk plan was documented clearly which ensured that everything on the right track.

By identifying all risks at the start, it helped the company to address potential issues before they get big which resulted in reducing the changes cost during working on the phases.

The project only got one error which is missing a functionality, but it did not have a large impact on the project. Also, the got some minor issues but the team members were able to track and solve these issues.

After finishing all the phases, they produced an enterprise system with a high quality as one of the project team members said “The performance of the ES, resulted in a very good acceptance of the ES by the end-users”

Overall, the waterfall approach provided a high-quality enterprise system for AirRoyal company.

Evaluateing how criteria like cost, time, resources, and legal requirements affect the decision-making process and the overall feasibility of the MomChef Homemade Bites App project.

The criteria cost, time, resources, and legal requirements have a big impact on the decision-making process through the project building, in this part I will discuss the positive and negative impact for each criteria in the MomChef Application.

For the cost, I tried to analyse the project and set a clear budget to Implement the project as good as possible based on the cost which helped me to identify most crucial feature. Also, setting a clear budget with calculating the return of investment helped me to convince stakeholders that the project worth the budget. While all these advantages are good, we have disadvantages such as, there were unexpected cost which made me let go of some non-important features of the system to make the set budget stays as it is which may damage the economic feasibility created if we didn't do that.

The resources' part is connected to the cost part, as we hired a skilled team that have more than 3 years of experience in the industry to produce a high-quality application, which resulted in higher the cost. What increased productivity was labelling all roles and responsibilities which improved the software quality.

The time criteria was the most important and hardest part to set, as we have a limited time to learn "Adalo" and to deliver the project within the time. But with Gantt chart made time management effective, as I scheduled the project timeline in a clear, and efficient way which motivated me to finish the project before the deadline, and it resulted in finishing the project before the deadline in 4 days. But in the scheduled time was not 100% clear, as some tasks delayed due to non-effective task management, and the most delayed part was learning "Adalo" which resulted in letting go to some features because I didn't have time to do them.

For legal, I made sure that all the legal laws and regulations in Jordan are followed, as I take all the required approvals to develop the MomChef Application which allowed us to be fully legal working application, because if we didn't follow these legal laws we might get charged and closes the application. Also, following all these legal laws made customer trust us more like the way customers prefer Uber than TaxiF because Uber follows the laws and TaxiF does not. But meeting all these legal laws needed more time, as we go to get the approval and we waited days to get the approval which we already handled in the timeline, as we put extra 2 days for each phase in case we didn't finish early.

Overall, it is very important to make sure that all these criteria are met, as they support the project success and affect the decision-making process during the project.

Justifying the use of data analytics, machine learning, and other data-driven approaches to enhance the performance, reliability, and user satisfaction of the MomChef Homemade Bites App.

At first, let's discuss what is data-driven. Data-driven is making decisions based on data analysis and interpretation. Data-driven have approaches that help companies to organize their data and ensure customer satisfaction to improve the overall performance. This includes approaches like data analytics, and machine learning, etc.

Data analytics is the process of collecting and analysing data related to the software to determine things that help in the decision-making process. In order to enhance the performance, reliability and user satisfaction of the MomChef Homemade app using data analytics, we could collect and analyse the peak ordering time, identify low ordering time, and who are the chefs that have the most orders. By analysing peak times, this would make the application to take more resources during these peak times, so we make sure that there are enough resources to keep the application run correctly and to prevent slowdowns or shutdowns which enhance the performance, reliability, and user experience. Also, by analysing low ordering time, we could put new discounts or offers to boost the ordering rate which boost the user satisfaction. Also, by analysing who are the chefs that have the most orders, we could make new category for best-seller chef, which help the users to allocate these chefs easily which boost the user experience and increase the user satisfaction.

Machine learning is part of the artificial intelligence, and a software that use artificial intelligence are more successful as it can be used in a lot of aspects of applications. In order to enhance the performance, reliability and user satisfaction of the MomChef Homemade app using Machine learning, we could use it with the data analytics. As we can analyse customers orders, as which are the meals that are most ordered by the customer then use machine learning to recommend meals that are like the meal the customer order the most, for example if there is a customer who usually order "Arabian food", we could use machine learning to recommend other "Arabian food" more than other types of food for the customer which enhance the user satisfaction rate. Also, machine learning could be used to predict failures, which help us to maintain the failure before it happens which enhance the reliability.

In conclusion, using data-driven approaches like data analytics and machine learning could enhance the performance, reliability, user satisfaction of the Momchef application which helps us to achieve our goals and objectives.

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